MILITARY SPECIFICATION

MANUAL KEYBOARD: 10526191

1. SCOPE

1.1 This specification covers one type of manual keyboard part of the Gun Direction Computer: M18. The output of this keyboard consists of 8 bit, digital coded, electrical signals for use in the computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-194	Painting and Finishing Systems for Fire Control Instruments
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10526191	Keyboard

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PACKAGING DATA SHEET

MIL-P-14232/P10526191 Packaging of Manual Keyboard

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.1.1 Use of non-standard parts and materials.- The use of a non-standard part or material is acceptable if a standard part is not in existance, or if its use will result in a substantial improvement in performance, reliability, simplicity of design or operation. When such a non-standard part is used, its use shall be approved by the procuring activity.

3.1.2 Connectors.- Connectors functioning as input/output (interface) terminations shall be as designated in Drawing 10526191.

3.1.3 Hardware.- Stainless steel fasteners (bolts, screws, nuts, washers, etc.) shall be utilized throughout the construction of the equipment.

3.1.4 Solid-state devices.- The equipment design shall make full use of solid-state devices, where their use will aid in accomplishing the requirements of weight, size, reliability, power, cooling, etc. However, the use of solid-state devices shall not alter the equipment in such a manner that it will not meet the requirements of this specification.

3.2 Design and construction.- The keyboard shall be designed and constructed in accordance with Drawing 10526191.

3.2.1 Fire hazard.- The design and construction of the equipment shall be such that its use shall not, under any conditions of storage or operation, create a fire hazard.

3.2.2 Special tools.- The equipment shall require no special tools for assembly, disassembly or maintenance.

3.2.3 Mounting attitude.- The equipment shall operate satisfactorily when mounted in any position.

3.2.4 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawings number requirements of MIL-D-70327 shall govern changes in the manufacture's part number.

3.2.5 Reference designations.- Reference designations shall be in accordance with MIL-STD-16. The unit numbering method described in MIL-STD-16 shall apply. The assignment of reference designations shall be made on schematic diagrams and drawings in conformance with MIL-D-70327.

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3.2.6 Finishing.- The finishing procedure shall conform to the general specification specified in 3.3.1 of the specification with the exception that paint color conforming to FED-STD-595 shall be applied to the following surfaces:

(a) Front and edges of front panel - color: olive drab, Color No. 34087 per FED-STD-595.

(b) Aluminum shall be anodized in accordance with MIL-STD-194. Where weldments are specified, the chrome acid finishes of MIL-STD-194 is required.

3.2.7 Keyboard.- The equipment shall consist of pushbuttons identified as illustrated in Drawing 10526191. The key cluster shall be centered and located within the outline dimensions shown in Drawings 10526191.

3.2.7.1 Color.- The keys shall be lusterless black, Color No. 37038 with permanent characters lusterless white, Color No. 37875 per FED-STD-595.

3.2.7.2 Shape.- The key shapes shall conform to the dimensions of Figure 1 and Table I. The key shall be suitably formed on the top surface, so as to be comfortable to the touch and to minimize any tendency of the fingers to slip from the key.

3.2.7.3 Nomenclature.- Key nomenclature shall be as listed in Table I.

3.2.7.4 Key travel.- The travel of each key shall be .187 $\pm$ .010 inches.

3.2.7.5 Key pressure.- The pressure required to operate each key shall be 13 $\pm$ 3 ounces throughout the temperature range of -40° Fahrenheit (F) to 135°F.

3.2.8 Pushbutton interlock.- The keyboard shall be so designed that:

(a) Normal finger pressure applied to any numbered pushbutton shall depress that pushbutton. When pressure is released, the pushbutton shall return to its normal (not depressed) position.

(b) When the enable line is energized, each depression of any key listed in Table II shall generate one and only one operation cycle.

(c) When any key listed in Table II is depressed prior to activating the enable line, one and only one operation cycle shall be generated when the enable line is activated. Farther de-activation and activation of the enable line shall not cause generation of an additional operation cycle until the key is released and depressed again.

(d) The keys shall be interlocked with each other in such a way as to assure that one and only one key may be actuated at a given time.

3.2.9 Wiring.- Wiring and connections shall be in accordance with Figure 2.

3.2.10 External connections.- Power and signal voltages will utilize a single cannon type DAM 25P electrical connector or equivalent. Cannot Type 20418-2 (2 required) locking devices or equivalent shall be used.

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3.2.11 Contact sealing.- All electrical contacts shall be enclosed to provide protection from dirt particles and other contaminants.

3.2.12 Voltage transient suppression.- Provisions shall be made to suppress voltage transients of inductive circuits and electrical arcing of contacts switching inductive loads within the equipment.

3.2.13 Weight.- The weight of the equipment shall not exceed 2.5 pounds.

3.3 General specifications.- The keyboard shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.3.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.3.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Handling
f. Etched circuit assembly (if circuit-boards are used)

3.4 Environmental service conditions.

3.4.1 Vibration.- The keyboard shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the keyboard for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the keyboard shall not be damaged or adversely affected and shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5.

3.4.2 Shock.- The keyboard shall withstand 3 shock impulses in each of 3 mutually perpendicular directions. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5.

3.4.3 Temperatures.

3.4.3.1 Operating.- The keyboard shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5 while exposed to temperatures of -40°F and +135°F.

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3.4.3.2 Storage.- The keyboard shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The keyboard shall meet the requirements of 3.2.7.5, 3.2.8 and 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.4.4.2 Non-operating.- The keyboard shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.4.5 Fungus resistance.- The keyboard shall show no evidence of damage and shall meet the requirements of 3.2.7.5, 3.2.8, and 3.5, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.4.6 Salt spray.- The keyboard shall show no evidence of physical damage and meet the requirements of 3.2.7.5, 3.2.8, and 3.5 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4.7 Sand and dust.- The keyboard, simulated in its computer mounting position, shall be capable of operation for 6 hours in an atmosphere heavily laden with sand and dust particles at an ambient temperature of 77° ± 15°F. It shall also be capable of operation after exposure for 6 hours to the sand and dust conditions at 160°F.

3.5 Performance.- Unless otherwise specified the keyboard shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

3.5.1 Pulse coding.- The selected electrical pulse code generated by actuation of a key shall be as shown in Table II. The symbol "X" associated with the key function indicated a -10 ± 3.0 Vdc pulse signal. All other information output signals shall be 0 ± .2 Vdc.

3.5.2 Contact rating.- If the keyboard utilizes mechanical contacts for the items identified as Kc1 through Kc8, Kcc, SM, and RECALL as shown in Figure 2, these contacts shall be capable of carrying all values of current within the range of 0 to 100 MA at 10 Vdc continuous duty.

3.5.3 SM and RECALL keys.- When the "SM" key is actuated and held in the activated position the voltage on pins 16 and 21 shall be -10 ± 3.0 Vdc, and 0 ± .2 Vdc at pins 18 and 19. When the "RECALL" key is actuated and held in the activated position the voltage at pins 18 and 19 shall be -10 ± 3.0 Vdc, and 0 ± .2 Vdc at pins 16 and 21.

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3.5.4 Contact timing.- The keyboard is required to provide an 8 bit parallel output level of approximately 25 milliseconds duration. Contact timing shall be such that the common output line will close after the information lines close, and return to normal before the information lines open.

3.5.4.1 Timing requirements for SM and RECALL keys.- Following depression of either the SM or RECALL key, the voltage at pin 18 or pin 21 respectively, shall change to 0 ± .2 Vdc at least 1 microsecond prior to a change of voltage on pin 16 or 19 respectively to outside the range of 0 ± .2 Vdc. Following the release of either the SM or RECALL key, the voltage on pin 16 or 19, respectively, shall change to 0 ± .2 Vdc at least 1 microsecond prior to a change of voltage on pin 18 or 21 respectively, to outside the range of 0 ± .2 Vdc.

3.5.4.2 Synchronizing signals (TC and TCK').- The "TC" signal at pin 9 (see Figure 3) shall be -10 ± 3.0 Vdc during the 25 ± 15 milliseconds (msec) true period and 0 ± .2 Vdc at all other times. The "TCK'" signal at pin 10 shall be 0 ± .2 Vdc during the false period and -10 ± 3.0 Vdc at all other times.

3.5.5 Signal bounce.- There shall be no signal bounce at the output TCK', pin 10 during the period when TC is true (see Figure 3).

3.5.6 Input power.- The following power levels are available for use in the keyboard. The equipment shall be designed to operate and maintain specified performance over all tolerance variations of voltages and under all operating conditions of the specification.

-31 ± 7 Vdc; maximum ripple 30% peak to peak;
maximum current 1.2 amperes peak under any operating temperature

-10 ± 1.0 Vdc; maximum current 12 milliamperes
for each output level energized
when loaded with 1K resistors.

-18 ± 1.8 Vdc; maximum current 50 milliamperes
for any operating conditions when
loaded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Table III, IV, V, & VI and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE III
(100 percent inspection)

Characteristics	Requirement	Test Procedures
101. Pushbutton interlock	3.2.8(a) to (d) incl.	4.5.2.3.10
. Pulse coding	3.5.1	4.5.2.3.2
102. Contact rating	3.5.2	4.5.2.3.3
103. SM and RECALL keys	3.5.3	4.5.2.3.4
104. Contact timing	3.5.4	4.5.2.3.7
105. Timing requirements for SM and RECALL keys	3.5.4.1	4.5.2.3.5
106. Synchronizing signals (TC and TCK')	3.5.4.2	4.5.2.3.6
107. Signal bounce	3.5.5	4.5.2.3.7

4.2.5 Certification provisions.- Acceptance inspection for material, design and construction in compliance with 3.1 and 3.2 shall be in accordance with 4.5.2.1.1.

4.3 Standard sampling.

4.3.1 General specifications.- One keyboard shall be selected at random by a Government representative as a standard sample from each 100 produced to determine compliance with 3.3. A visual examination shall be conducted to determine conformance with MIL-F-13926 and MIL-C-46357.

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4.3.2 Materials and construction.- Three units shall be selected at random by a Government representative as standard samples from each 50 produced or from each month's production, whichever occurs first. Examination, of the standard samples, shall be as specified in 4.5.2.1.2 for each characteristic listed in Table VI.

4.3.3 Environmental.- Three units shall be selected at random by a Government representative as standard samples from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table IV.

TABLE IV

Characteristic	Requirement	Test Procedures
301. Vibration	3.4.1	4.5.1.1
302. Shock	3.4.2	4.5.1.2
303. Humidity-Operating	3.4.4.1, 3.2.7.5 3.2.8, 3.5	4.5.1.3.2, 4.5.2.2, 4.5.2.3
304. Humidity-Non-operating	3.4.4.2	4.5.1.3.1
305. Salt spray	3.4.6, 3.2.7.5, 3.2.8, 3.5	4.5.1.4, 4.5.2.2, 4.5.2.3
306. Operating temperature	3.4.3.1, 3.2.7.5, 3.2.8, 3.5	4.5.1.5, 4.5.2.2, 4.5.2.3
307. Storage temperature	3.4.3.2	4.5.1.5
308. Sand and dust	3.4.7, 3.2.7.5, 3.2.9, 3.5	4.5.1.7, 4.5.2.1.2, 4.5.2.2 & 4.5.2.3

4.3.4 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.4.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.4.1 with the keyboard rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with functional requirements specified under 3.4.1.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.4.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5 G's. The keyboard shall be subsequently inspected for physical damage or defects in compliance with 3.4.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the functional requirements required by 3.4.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.4.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.4.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 $\pm$ 5 percent for the 24 hour period and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.1.2, 4.5.2.2, and the requirements under 4.5.2.3 to determine compliance with 3.4.4.1.

4.5.1.4 Salt spray.- The keyboard shall be subjected to the conditions in 3.4.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.1.2, 4.5.2.2, and the requirements under 4.5.2.3 to determine compliance with 3.4.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the keyboard shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.4.3 shall apply. While at the operating temperatures of 3.4.3.1, the unit shall be tested in accordance with 4.5.2.1.2, 4.5.2.2 and 4.5.2.3 to determine compliance with the requirements of 3.4.3 as required by 3.4.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.1, 4.5.2.2 and the requirements under 4.5.2.3 to determine compliance with 3.4.3.2.

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4.5.1.6 Fungus.- To determine compliance with 3.4.5, one of the first 5 keyboards produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.1.7 Sand and dust.- This test shall be conducted in accordance with the sand and dust test defined in MIL-F-13926 to determine compliance with 3.4.7. Temperature changes shall not occur at a rate exceeding 2 degrees Fahrenheit per minute to avoid thermal shock. The keyboard shall be positioned in the same manner as it is positioned when functioning as a part of Computer Set, Gun Direction: M18. Operation while at 77°F shall be conducted in accordance with the tests in 4.5.2.1.2, 4.5.2.2 and the requirements under 4.5.2.3.

4.5.2 Performance.

4.5.2.1 Materials and construction.

4.5.2.1.1 Certification.- Certification shall include test data and results for all characteristics specified in Table V. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor and Government approval therefore.

TABEL V - CERTIFICATION

Characteristics		Requirement
301.	Materials	3.1
302.	Non-standard parts and materials	3.1.1
303.	Connectors	3.1.2
304.	Hardware	3.1.3
305.	Solid-state devices	3.1.4
306.	Fire hazard	3.2.1
307.	Special tools	3.2.2
308.	Interchangeability	3.2.4
309.	Finishing	3.2.6(b)
310.	External connections	3.2.10
311.	Voltage transient suppression	3.2.12

4.5.2.1.2 Examination.- In addition to measurements by Standard Measuring Equipment (SHE), a visual and tactile examination shall be made for all requirements listed in Table VI below:

TABLE VI

Characteristics		Requirement	Examination
401.	Mounting attitude	3.2.3	Visual, Tactile, SME
402.	Reference designations	3.2.5	MIL-STD-16, MIL-D-70327, Visual
403.	Finishing	3.2.6(a)	FED-STD-595, Visual
404.	Keyboard	3.2.7	Visual, SME

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Characteristics	Requirement	Examination
405. Color of keys	3.2.7.1	FED-STD-595, Visual
406. Color of characters	3.2.7.1	FED-STD-595, Visual
407. Shape	3.2.7.2	Visual, Tactile, SME
408. Nomenclature	3.2.7.3	Visual
409. Key travel, Pushbutton interlock	3.2.7.4, 3.2.8(a)	SME, Tactile
410. Wiring	3.2.9	Visual
411. Contact sealing	3.2.11	Visual
412. Weight	3.2.13	SME

4.5.2.2 Key pressure.- Test equipment to measure direct force of the pressure or push (in ounces) shall be within the range of zero to twenty ounces. The accuracy of the measurement shall be plus or minus one-half ounce throughout the entire zone of operation. The pressure required to operate each key from their normal position shall not exceed the limit specified in 3.2.7.5 for the temperature indicated.

4.5.2.3 Electrical tests.- Each completely assembled keyboard shall be subjected to the following tests:

4.5.2.3.1 External loading.- Connect a 1000 ohm ± 5%, 1 watt resistor to each of the following pins: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 16, 18, 19 and 21. Return other side of above resistors to circuit ground.

4.5.2.3.1.1 Input power application.- Apply the following voltages to the specified pins 6 permit testing with input power specified in 3.5.6.

-31 ± 7 Vdc; pin 12
-10 ± 1.0 Vdc; pin 11
- 1 ± 0.1 Vdc; pin 13
-18 ± 1.8 Vdc; pin 23
Ground; pins 14 and 15

4.5.2.3.2 Pulse coding.- The selected electrical pulse code generated by actuation of a key shall be as shown in Table II. The symbol "X" in Table II associated with the key function indicates a -10 ± 3.0 Vdc level, 10 to 50 millisecond duration to be measured across the corresponding resistor loads. All other information outputs shall measure 0 ± .2 Vdc as measured with standard measuring equipment to determine conformance with 3.5.1.

4.5.2.3.3 Contact rating.- If the keyboard utilizes mechanical contacts for pins Kc1 through Kc8, Kcc, SM, and RECALL as shown in Figure 2, these contacts when measured with standard measuring equipment shall verify the current capacity as specified in 3.5.2.

4.5.2.3.4 SM and RECALL keys.- Each SM and Recall key controls a single pole double throw switch or equivalent. With the "SM" key actuated and held in the activated position, a -10 ± 3.0 Vdc level shall be measured on activated position, a -10 ± 3.0 Vdc on pins 16 and 21, and 0 ± .2 Vdc on pins 18 and 19, With the "Recall" key actuated and held in the activated position, a -10 ± 3.0 Vdc level shall be measured at pins 18 and 19, and 0 ± .2 Vdc at pins 16 and 21 when measured with standard measuring equipment to verify conformance with 3.5.3.

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4.5.2.3.5 Contact timing requirements for SM and RECALL keys.- Following depression of either the SM or RECALL key, the voltage at pin 18 or pin 21 respectively, shall change to 0 ± .2 Vdc at least 1 microsecond prior to a change of voltage on pin 16 or 19 respectively to outside the range of 0 ± .2 Vdc. Following the release of either the SM or RECALL key, the voltage on pin 16 or 19, respectively, shall change to 0 ± .2Vdc at least 1 microsecond prior to a change of voltage on pin 18 or al respectively, to outside the range of 0 ± .2 Vdc as measured with standard measuring equipment to verify conformance with 3.5.4 and 3.5.4.1.

4.5.2.3.6 Synchronizing signals (TC and TCK').- As viewed on the standard measuring equipment utilized, the signal TC at pin 9 shall be -10 ± 3.0 Vdc during the 25 ± 15 millisecond sampling period and 0 ± .2 Vdc at all other times. The signal TCK' at pin 10 shall be 0 ± .2 Vdc during its sampling period and -10 ± 3.0 Vdc at all other times to determine conformance with 3.5.4.2.

4.5.2.3.7 Signal bounce and contact timing.- There shall be no signal bounce at the output TCK', pin 10 during the period when TC is true as viewed on the standard measuring equipment utilized, (see Figure 3) to determine conformance with the requirements of 3.5.4 and 3.5.5.

4.5.2.3.8 Input power.- Unless otherwise specified herein all keyboards shall be tested with the power sources as defined in 3.5.6.

4.5.2.3.9 Disable test.- Remove the -1 ± .1 Vdc from pin 13. Connect an external resistor between circuit ground and pin 13 with a resistance of proper value to produce a current of 5 ± 0.5 milliamperes flowing into the keyboard. Under this conditon, there shall be no pulse codes generated on any of the output pins 1 through 10 when any of the keys are pressed. This test will verify compliance with 3.2.8 (b) and (c).

4.5.2.3.10 Pushbutton interlock (Electrical).- Depress any key listed in Table II and hold depressed for the remainder of this test. Remove the resistance specified in 4.5.2.3.9 from pin 13 and connect -1 ± .1 Vdc to pin 13. Observe that the keyboard generates one and only one operation cycle. Repeat the cycle several times of alternately connecting the resistance and -1 ± .1 Vdc and observe that no additional operation cycles are generated to determine conformance with 3.2.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging DataSheet MIL-P-14232/P10526191.

6. NOTES

6.1 Intended use.- The keyboard shall be used to manually program binary coded decimal information into the Gun Direction Computer: M18.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable Packaging Data Sheet number (see 5.1).
(e) A block of serial numbers to identify each keyboard.

Custodian:

Army - MU

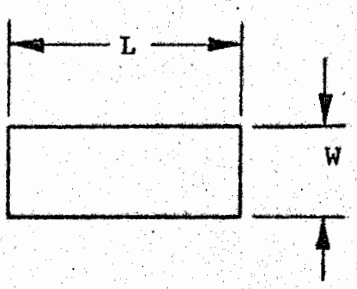
Preparing activity:

Army - MU

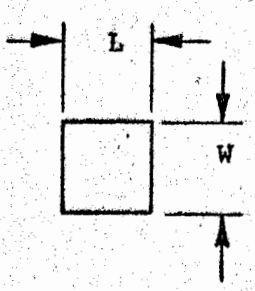
Project No. 1220-A195Y

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Check the source to verify that this is the current version before use.

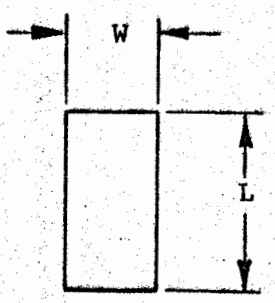
KEY NOMENCLATURE	KEY ORIENTATION		NOTES
	LENGTH	WIDTH	
0	1-1/4	1/2	TYPE I
1	1/2	1/2	TYPE II
2	1/2	1/2	TYPE II
3	1/2	1/2	TYPE II
4	1/2	1/2	TYPE II
5	1/2	1/2	TYPE II
6	1/2	1/2	TYPE II
7	1/2	1/2	TYPE II
8	1/2	1/2	TYPE II
9	1/2	1/2	TYPE II
•	1/2	1/2	TYPE II
CLEAR	1/2	1/2	TYPE II
ENTER	1/2	1/2	TYPE II
LEFT DOWN DROP -	7/8	1/2	TYPE III
RIGHT UP ADD +	7/8	1/2	TYPE III
SM	7/8	1/2	TYPE III
RECALL	7/8	1/2	TYPE III



TYPE I



TYPE II



Type III

TABLE I

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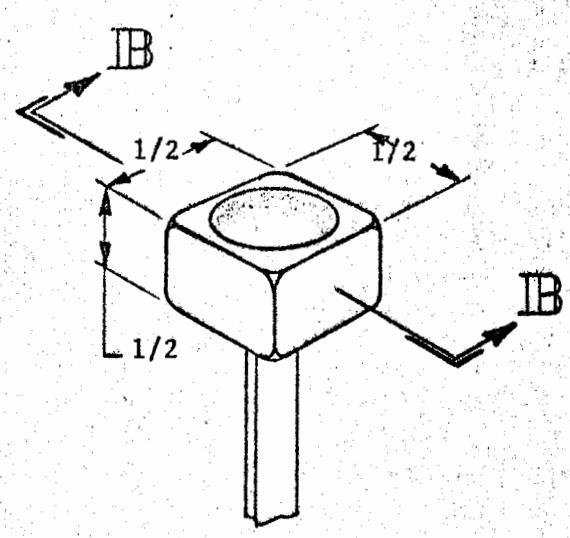
KEY FUNCTION	CODE							
	8	7	6	5	4	3	2	1
0			X	X				
1	X		X	X				X
2	X		X	X			X	
3			X	X			X	X
4	X		X	X		X		
5			X	X		X		X
6			X	X		X	X	
7	X		X	X		X	X	X
8	X		X	X	X			
9			X	X	X			X
LEFT DOWN DROP -			X					X
RIGHT UP ADD +			X				X	
DECIMAL POINT	X		X	X	X	X		X
CLEAR	X	X		X	X	X		X
ENTER		X				X		

X indicates a true condition, i.e., output required on corresponding information lines.

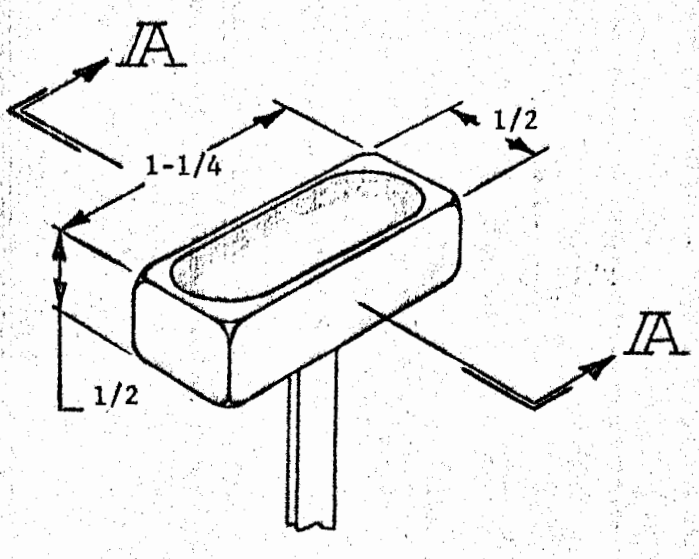
NOTE: The "SM" and "Recall" keys each activate a SPDT switch or equivalent. The operation shall be as specified in 3.5.3.

TABLE II

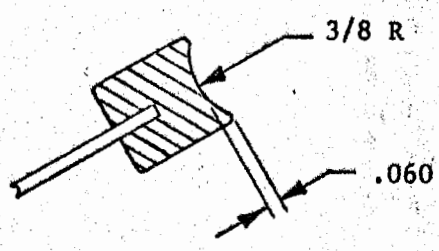
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.



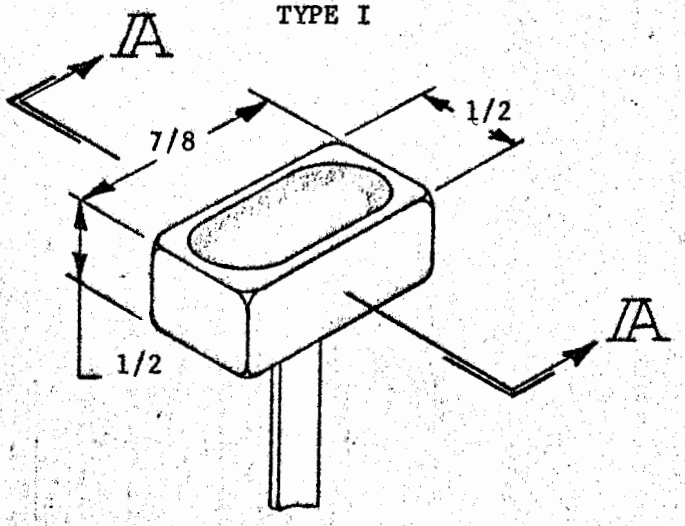
TYPE II



TYPE I

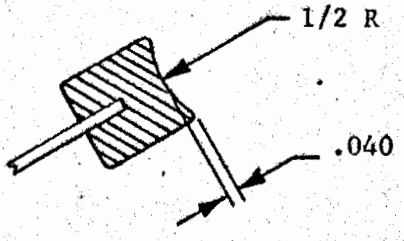


SECTION B-B



TYPE III

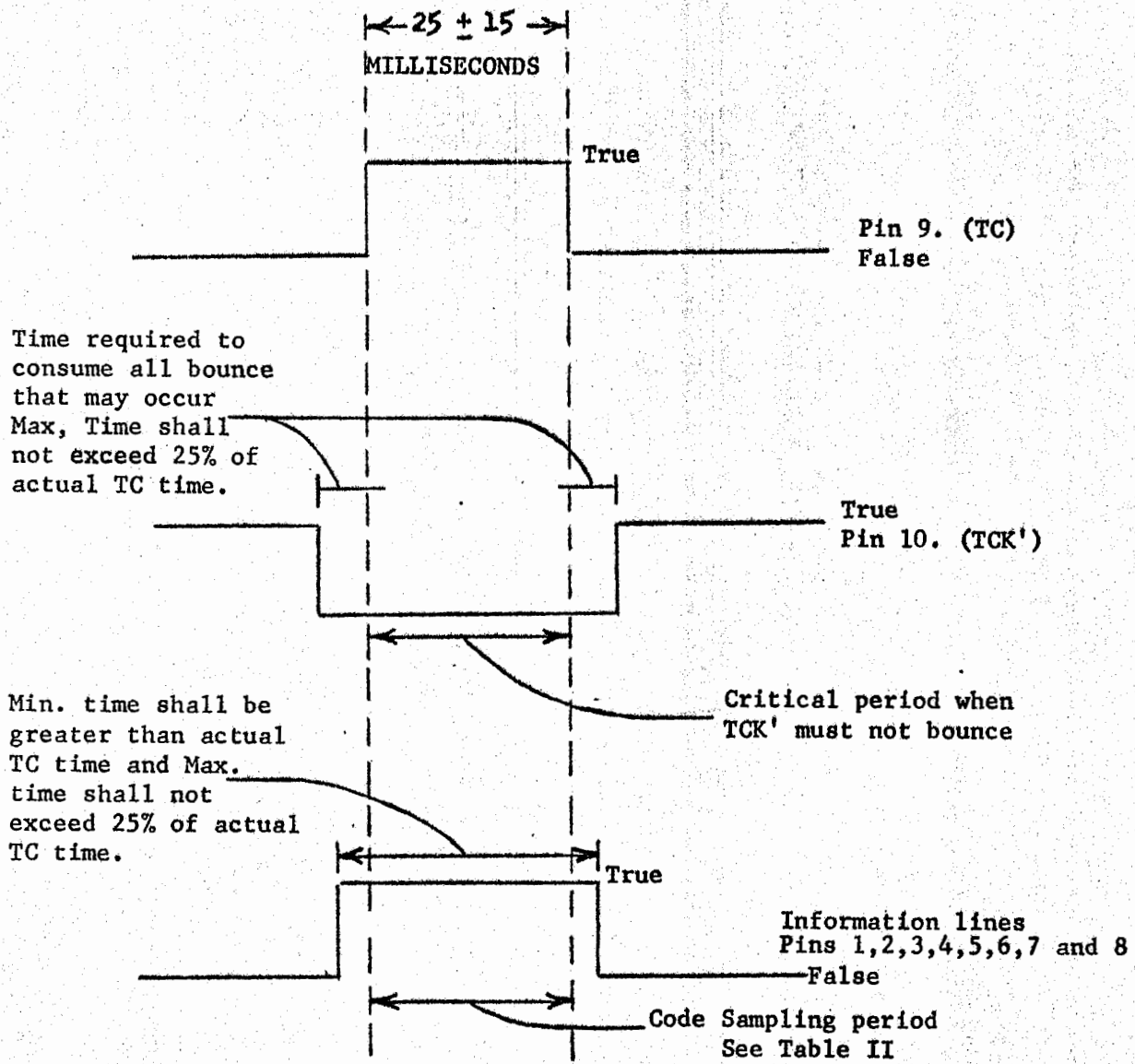
NOTE: All corners and edges to have 1/32R. (typical).



SECTION A-A

FIGURE 1

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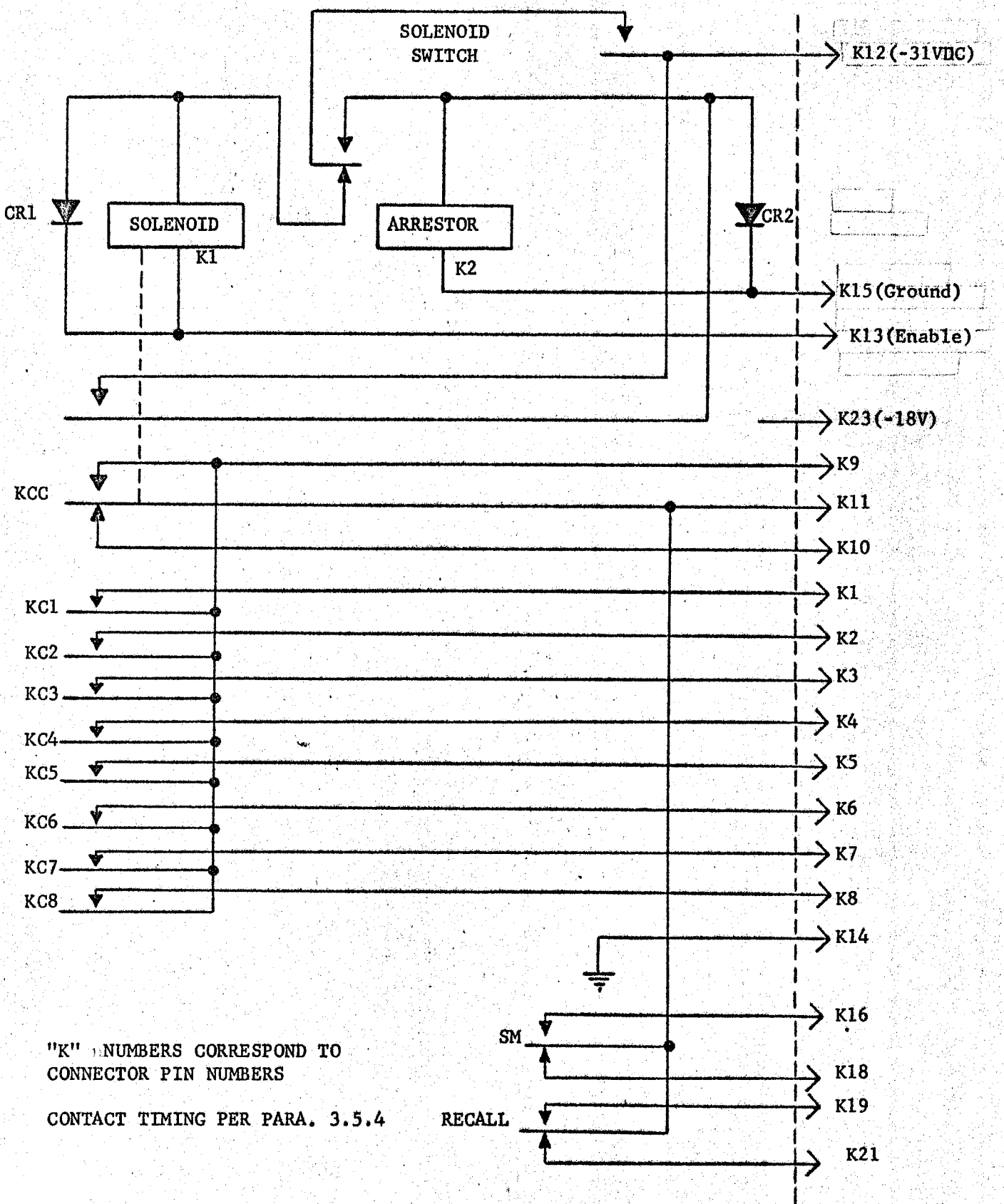
True - Indicates a closed circuit condition at the corresponding pin numbers.

False - Indicates an open circuit condition at the corresponding pin numbers.

FIGURE 3

Timing Sequence Chart

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"K" NUMBERS CORRESPOND TO CONNECTOR PIN NUMBERS

CONTACT TIMING PER PARA. 3.5.4

FIGURE 2

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